

**UNIVERSITY OF MPUMALANGA**

School of Computing and Mathematical Sciences

Diploma in Information and Communication Technology

**DICT421: Advanced Application Development**

**PRACTICAL:** Concept of Branching in Git

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Suppose you are working on an application that reads student scores and then assigns grades based on the following scheme:

- i) Grade is A if score is greater than or equal to 85 and less than or equal to 100;
- ii) Grade is B+ if score is greater than or equal to 75 and less than 85;
- iii) Grade is B if score is greater than or equal to 70 and less than 75;
- iv) Grade is C if score is greater than or equal to 60 and less than 70;
- v) Grade is D if score is greater than or equal to 50 and less than 60;
- vi) Grade is F otherwise.

The program prompts the user to enter the total number of students and then prompts the user to enter all of the scores and concludes by displaying the grades.

You would like to add a new feature to the application, i.e., a feature to get the best score and display the best score using the concept of branching in Git.

**Hint:** Stages to follow:

1. Create a Git repository on your local machine
2. Create the application, add and commit it to the Git repository
3. Creating a new-feature branch from the master branch of the Git repository
4. Add the new feature to the application in the new-feature branch
5. Once the new feature to the application has been completed and tested, add and commit the updated application to the new-feature branch
6. Merge the new-feature branch into the master branch
7. Delete the new-feature branch since it is no longer needed.
8. Push the master branch to a remote repository, i.e. GitHub.

Show your work by taking screenshots at each stage.